

Digital Lending: Portfolio Optimization

Credit Policy Recommendation Report

10,000

Loans Analysed

42.5%

Portfolio Default Rate

3,488

NTC Customers

0.6985

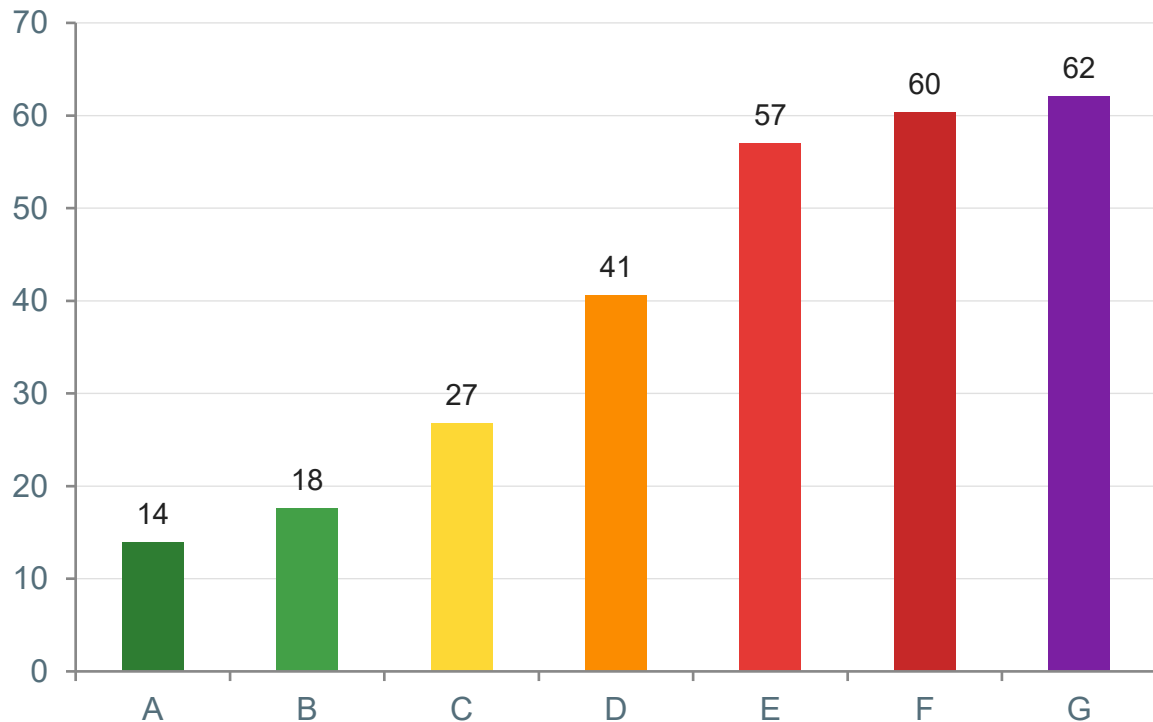
Best Model AUC

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CA Club, IIT Guwahati · To: Ms. Priya Sharma, CRO · June 2026

Default Rate by Loan Grade

The steepest risk gradient in the portfolio — Grade E–G default at 4× the rate of Grade A–B

Default Rate by Loan Grade (%)



Grade A–B

14–18%

Approve freely

Grade C–D

27–41%

Conditional approval

Grade E–G

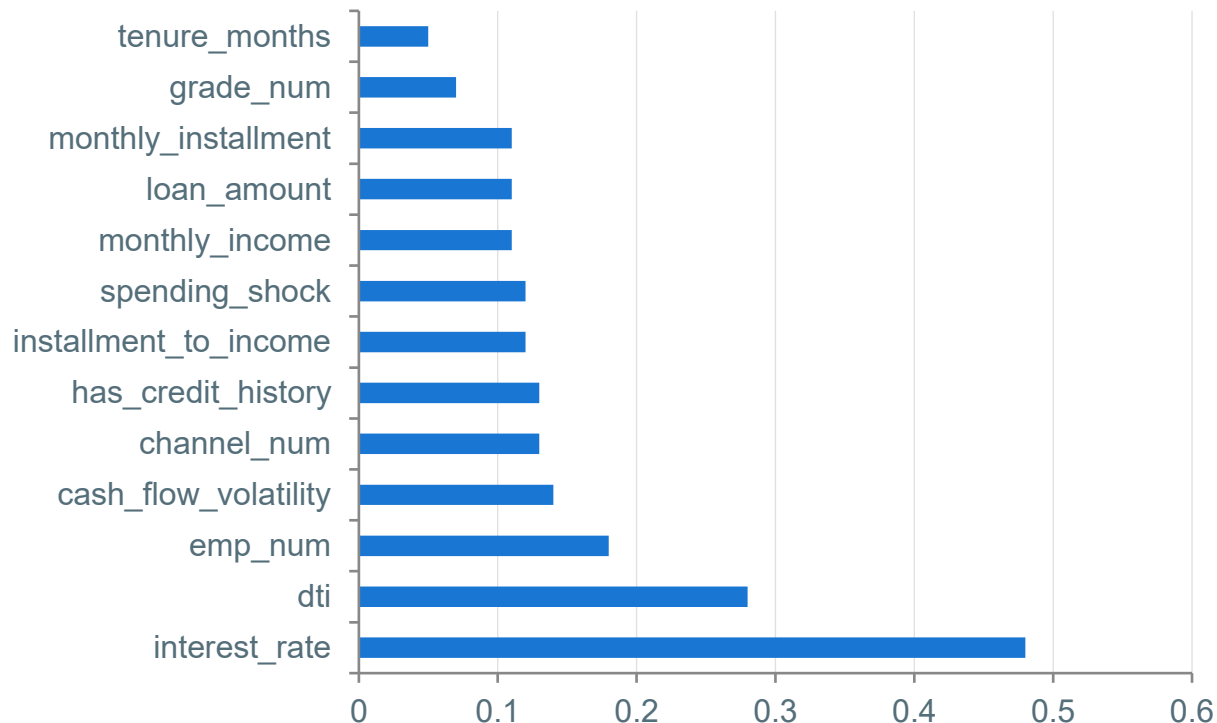
57–62%

Restrict / Decline

SHAP Analysis — Top Default Predictors

Interest rate and DTI dominate; together they account for the majority of model predictive power

Mean |SHAP Value| — Average Impact on Default Prediction



0.699

Logistic Regression AUC

0.693

XGBoost AUC

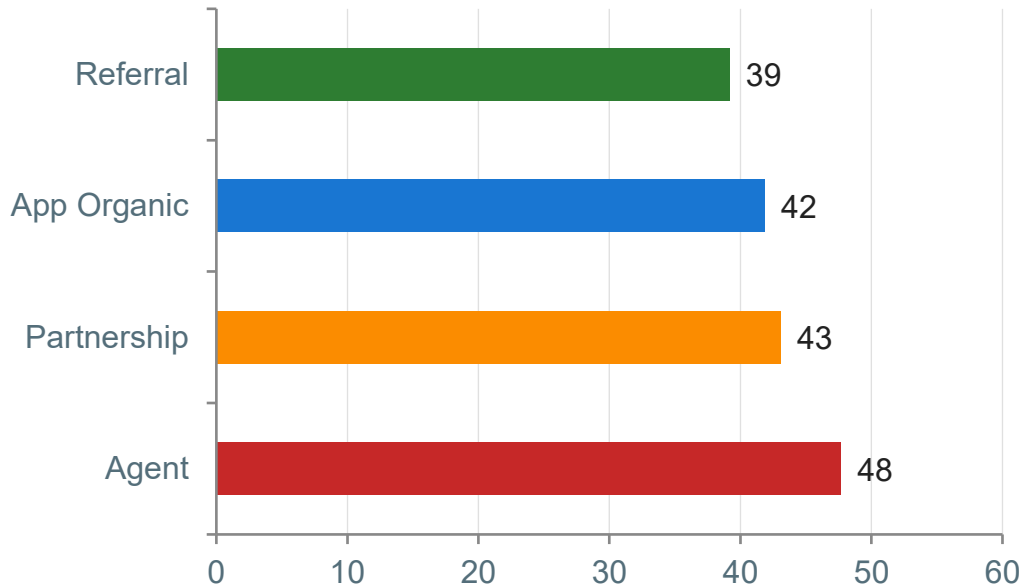
The Rate-Trap

Interest rate is the #1 predictor. Higher rates → higher DTI → higher default → even higher rates assigned. Breaking this cycle with tiered rate caps for Grade D borderline borrowers is the single highest-impact policy lever.

Acquisition Channel Analysis

Channel quality reflects borrower screening integrity — evaluators asked specifically about this

Default Rate by Acquisition Channel (%)



8.5pp spread between best (Referral 39.2%) and worst (Agent 47.7%) — addressable through underwriting reforms.

Referral 39.2%

Incentivize — 0.5% rate discount. Referrer accountability clause.

App Organic 41.9%

Maintain — primary growth channel. Improve UX for Grade A–C conversion.

Partnership 43.1%

Monitor — quarterly portfolio audit. Renegotiate if DR > 45%.

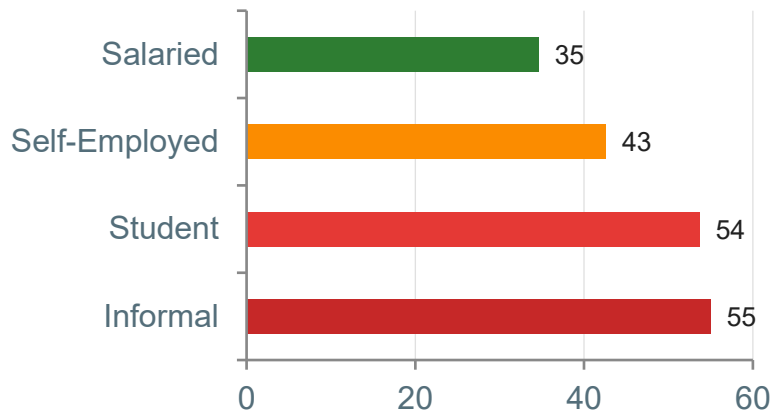
Agent 47.7%

Tighten — mandatory income verification. Deactivate agents with DR > 55%.

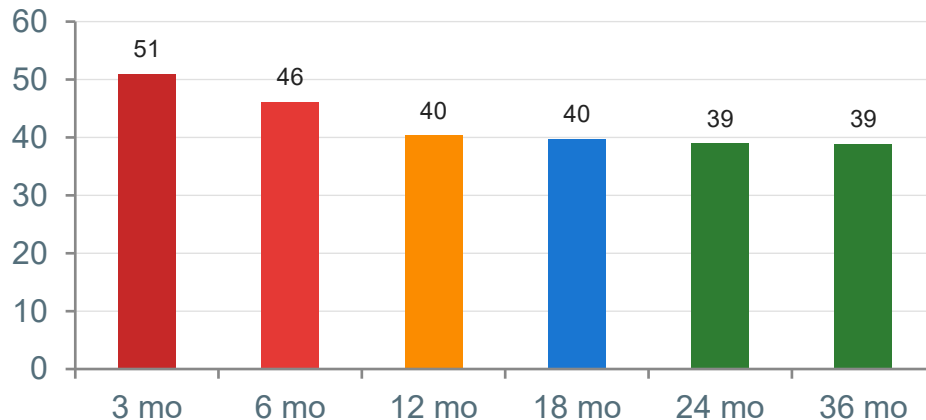
Employment Type, Loan Size & Tenure Risk

Employment drives 20pp default spread; tenure direction is counterintuitive

Default Rate by Employment Type (%)



Default Rate by Tenure (%) — Shorter ≠ Safer



Salaried = sweet spot
34.6% default rate
Largest volume (45%)

3-month tenures risky
50.9% default rate
Concentrated cash stress

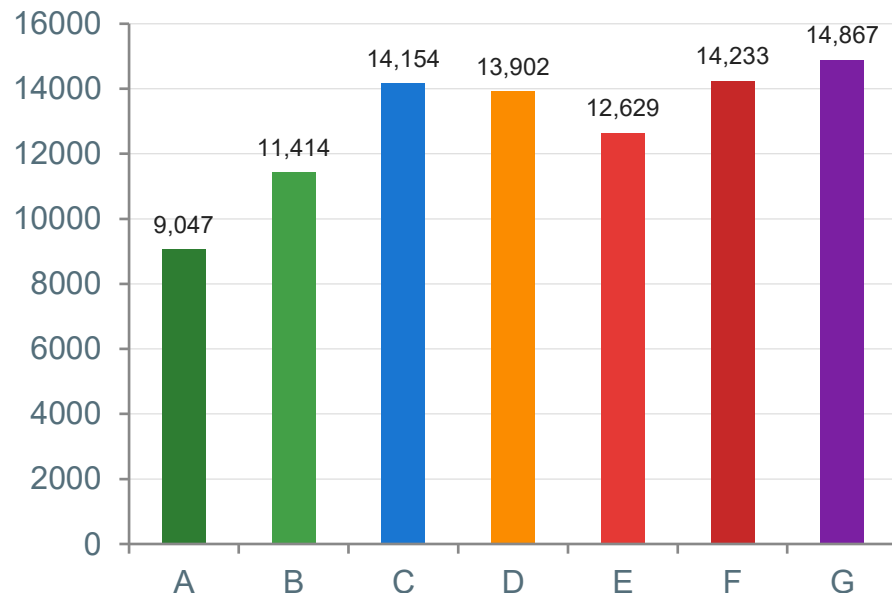
Grade D–E: recommend
12–18 month products
Balance exposure vs burden

Gap Analysis & CLV — Where Value Lives

Grade C maximises risk-adjusted revenue; defaulters worse across every financial metric

Metric	Fully Paid	Defaulted	Δ
DTI	0.47	0.58	+23%
FICO Score	695.7	669.1	-3.8%
Installment/Income	0.29	0.40	+38%
Monthly Income	₹36,794	₹30,935	-16%
Loan Amount	₹73,668	₹78,000	+5.9%
Interest Rate	22.3%	25.5%	+3.2pp

Expected Revenue per Loan After Default (₹)



Grade C = best expected revenue (₹14,154) at 26.8% default → optimal growth target

New-to-Credit (NTC) Customer Policy

3,488 customers (34.88%) have no credit history — exclusion forfeits a major growth opportunity

3,488

NTC Customers

34.88%

Portfolio Share

FICO Unavailable

Standard scoring excludes them entirely

NTC Qualifying Criteria (all 3 must pass)

1

Installment-to-Income Ratio

< 0.35

Repayment burden stays manageable without bureau history

2

Debt-to-Income (DTI)

< 0.50

Caps total debt obligation at safe level for unproven borrowers

3

Employment Type

Salaried or Self-Employed
only

Informal workers & Students excluded — income too unstable

Starter Loan: ₹30,000 cap, 12-month tenure → 2 successful repayments → Graduate to standard grade framework (high LTV, low acquisition cost)

Early Warning Signal System

3-tier SHAP-driven framework to detect financial stress before default occurs

WATCH

Trigger: $DTI > 0.50$ OR $spending_shock = 1$

Action: Flag for monitoring; send repayment reminder

Review:
Monthly

ALERT

Trigger: $DTI > 0.55$ AND ($installment_to_income > 0.35$ OR $cash_flow_volatility > 0.75$)

Action: Proactive outreach; offer loan restructuring options

Review:
Bi-weekly

CRITICAL

Trigger: $Grade \geq E$ AND $DTI > 0.55$ AND $spending_shock = 1$

Action: Immediate intervention; escalate to collections team

Review:
Weekly

Rate-Trap Fix: Grade D FICO 665–680 → offer –1.5% rate for auto-debit enrollment → breaks the high-rate → high-default → higher-rate feedback cycle

Credit Policy Recommendations

Grade-based approval framework with employment, channel, and NTC guardrails

Grade	Decision	Max Loan	Tenure	Conditions	Risk Tier
A	APPROVE	No cap	36 mo	Standard; loyalty pricing eligible	Low
B	APPROVE	₹3,00,000	36 mo	Standard terms	Low
C	APPROVE	₹75,000	24 mo	Rate ceiling 20%; loan cap enforced	Medium
D	CONDITIONAL	₹50,000	18 mo	Income proof required; DTI < 0.45	High
E	RESTRICTED	₹25,000	12 mo	Salaried only; credit history mandatory	Very High
F	DECLINE	N/A	N/A	Only if Salaried + FICO > 640	Critical
G	DECLINE	N/A	N/A	Decline all applications	Critical

Key Guardrails:

DTI hard cap: 0.65

Installment/Income $\leq$ 40%

Spending shock → manual review

Grade D FICO 665–680: –1.5%
rate for auto-debit

NTC Policy (Section 4.5): 3,488 customers without credit history → qualify via ITI < 0.35, DTI < 0.50, Salaried/SE only → ₹30K starter loan → graduate to grade framework after 2 repayments

Projected Impact & Leadership Dashboard

3-lever policy change projected to nearly halve the default rate

Default Rate

42.5%

→ **~27%**

Grade F/G Loans

865

→ **~100**

Agent DR

47.7%

→ **~40%**

Grade A–C Share

~25%

→ **>40%**

Monthly Leadership KPI Dashboard

KPI Metric	Current	Target	Alert
Portfolio Default Rate	42.5%	< 30%	> 35%
Grade A–C Approval Share	~25%	> 40%	< 30%
Agent Channel Default Rate	47.7%	< 42%	> 50%
DTI > 0.5 in New Approvals	~35%	< 20%	> 30%
Salaried Borrower Share	~45%	> 55%	< 40%
30-Day Delinquency Rate	—	< 15%	> 20%

Conclusion

- 01 Dataset note: 42.5% default rate is intentional — stress-test design to enable policy framework for all risk segments including Grade E–G
- 02 Grade C is the optimal growth target — best expected revenue (₹14,154) at 26.8% default; Grades F–G add risk but minimal incremental revenue
- 03 Salaried + Referral = best segment-channel combination — lowest risk, highest volume, most scalable growth pathway
- 04 Interest rate is the #1 SHAP predictor — tiered rate caps for Grade D borderline borrowers break the default-rate feedback loop
- 05 3,488 NTC customers (34.88%) require alternative scoring — ITI/DTI/employment criteria enable inclusive policy without sacrificing quality

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Consulting & Analytics Club, IIT Guwahati · Digital Lending: Portfolio Optimization · June 2026